MD. Mehedi Hasan Maruf

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Summary

Software engineer with 3+ years of experience in full-stack web development (TypeScript, Python) and hands-on .NET desktop work. I built PharmaC, a WPF pharmacy management system on .NET 8 with Entity Framework Core and MVVM. At Code Studio I shipped Django and Node backends and ran their Docker deployments, CI/CD pipelines, and production Linux servers. I also do applied ML research: my B.Sc. thesis at Rajshahi University of Engineering & Technology (RUET) finds the university courses that hold students back, and a paper from it is under review at LAK 2027.

Experience

Software Engineer II

Jan 2025 – Present

Aamira Software Solutions, Kokomo, Indiana, USA (remote)

- Raised client response rates by 30% by leading development of a CRM system.
- Cut API response time by 40% by designing and building a TypeScript microservice backend.
- Made the application 60% faster by optimizing its Supabase (PostgreSQL) queries.
- Building a retrieval-augmented generation (RAG) feature for the platform.

Junior Software Engineer

Nov 2022 – Apr 2024

Code Studio, Rajshahi, Bangladesh

- Kept 12 Linux servers at 99.8% uptime, and cut deployment time by 40% by reworking the CI/CD pipelines.
- Moved 5 client environments onto Docker, which made them simpler to deploy.
- Wrote JavaScript, TypeScript, and Python (Django) backends used by thousands of people a day.

Industrial Trainee

Mar 2025

Brain Station 23, Dhaka, Bangladesh

- Represented a 15-person trainee cohort to management.
- Built and deployed StudentAI (see Projects), an AI learning assistant on FastAPI and Next.js.

Skills

AI/ML: LLM applications, RAG, LangChain, Hugging Face Transformers, PyTorch, NLP, vector search (pgvector), model evaluation (RLHF)
Languages: C#, TypeScript, Python, JavaScript
.NET: .NET 8, WPF, Entity Framework Core, MVVM
Backend: NestJS, FastAPI, Fastify, tRPC, Django
Frontend: Next.js (React), Angular
Databases: PostgreSQL (Supabase), MongoDB, MySQL
DevOps: Docker, GitHub Actions (CI/CD), Linux server administration

Selected Projects

PharmaC | Desktop Pharmacy Management System

Feb 2025

- Built a WPF desktop app (.NET 8, MVVM, WPF-UI) for pharmacy inventory, sales, and orders.
- Modeled medicines, brands, generics, and admin accounts in Entity Framework Core (SQL Server, MySQL).
- Implemented JWT authentication with token refresh, barcode lookup for medicine scanning, and per-device licensing from a CPU/motherboard/BIOS hardware fingerprint.
- Connected the app to a REST API (`HttpClient`) for medicines, orders, carts, and licenses, so inventory syncs in real time and checkout runs in BDT.

StudentAI | Intelligent Learning Companion *Mar 2025*

- An AI learning assistant (FastAPI, Next.js) built during industrial training at Brain Station 23.
- Answers questions about uploaded documents through a LangChain RAG pipeline with semantic retrieval and conversation memory; it also analyzes YouTube lectures and writes quizzes.
- Accepts PDFs, images, and video, and has real-time WebSocket chat, JWT login, and progress tracking.
- Deployed on Vercel and Render with a Supabase database.

Education

B.Sc. Engineering, Computer Science and Engineering *Mar 2022 – Aug 2026*

Rajshahi University of Engineering & Technology (RUET), Rajshahi, Bangladesh

- CGPA: 3.75 / 4.00, Honours
- Thesis: *TRACE*, which ranks the courses that hold students back using prerequisite graphs, Bayesian fail-rate models, and student-feedback sentiment, then checks the result with causal inference on 23,369 student records. A follow-up paper is under review at LAK 2027.

Certifications & Achievements

- Digital Accessibility Training (DAT) — a2i (Aspire to Innovate), Govt. of Bangladesh (Oct 2025)
- Industrial Training Certificate — Brain Station 23 (Mar 2025)
- PHP (Laravel) — EDGE Project (Feb 2025)
- Experience Certificate — Code Studio (Apr 2024)
- 4th Place — RUET CodeSmash, team competitive programming contest (Mar 2023)
- 4th Place — RUET CodeSpark, individual competitive programming contest (Oct 2022)